

IMAGE PROCESSING TECHNIQUES

Extended Class Activities Worksheet | UG Level | Activity 5: Thresholding Techniques

PART 1: THRESHOLDING TECHNIQUES COMPARISON

Technique	Used When	Advantage
Global Thresholding	Images have uniform, controlled background illumination and distinct bimodal intensity distributions.	Computationally simple, extremely fast, and easy to implement.
Adaptive Thresholding	Images have non-uniform (uneven) illumination, shadows, or varying background contrast across different regions.	Handles local lighting variations dynamically by calculating threshold values locally for small sub-regions.
Otsu's Method	Bimodal histogram images where an optimal global threshold value needs to be determined automatically without manual tuning.	Completely automatic, parameter-free, and mathematically optimizes class separation (maximizes inter-class variance).

PART 2: WHY GLOBAL THRESHOLDING FAILS UNDER UNEVEN ILLUMINATION

Global thresholding applies a single fixed intensity threshold T across the entire image. When illumination is non-uniform (e.g., due to shadows, gradients, or vignetting), the background intensity in brightly lit regions can become higher than the foreground object intensity in shadowed regions.

As a result, no single global value T can correctly separate object pixels from background pixels image-wide:

- Setting T too high misclassifies foreground objects in shadowed areas as background.
- Setting T too low misclassifies brightly illuminated background areas as foreground objects.